

22/04/2025

CODE-A



Aakash

Medical | IIT-JEE | Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015



AIM - 720

(Advanced INTENSIVE Mastery for 720)

MM : 720

CST-9

Time : 180 Mins.

Complete Syllabus of NEET

Instructions:

- (i) Duration of Test is 3 hrs.
- (ii) The Test consists of 180 questions. The maximum marks are 720.
- (iii) There are three parts in the question paper consisting of Physics, Chemistry and Biology having 45 questions each in Physics and Chemistry and 90 questions in Biology).
- (iv) Each question carries +4 marks. For every wrong response, -1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
- (v) Use blue/black ballpoint pen only to darken the appropriate circle.
- (vi) Mark should be dark and completely fill the circle.
- (vii) Dark only one circle for each entry.
- (viii) Dark the circle in the space provided only.
- (ix) Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

6 Aakashians scored a perfect 720/720 from Classroom Programs



720/720 **Mrdul M Anand**
3 Year Classroom



720/720 **Ayush Naugrallya**
4 Year Classroom



720/720 **Arghyadeep Dutta**
2 Year Classroom



720/720 **Aryan Yadav**
1 Year Classroom



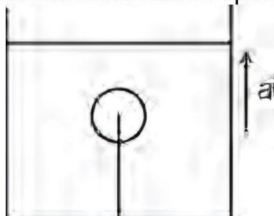
720/720 **Palansha Agarwal**
2 Year Classroom



720/720 **Iram Quazi**
1 Year Classroom

PHYSICS

1. A solid sphere of mass m and density ρ is held stationary relative to a tank filled with liquid of density d . The tank is accelerated with acceleration a . The tension in thread connected between sphere and bottom of tank would be



- (1) $m(g+a)\left(\frac{d}{\rho}-1\right)$
- (2) $m\left(\frac{\rho g}{d}-g-a\right)$
- (3) $m\left(\frac{dg}{\rho}+g+a\right)$
- (4) Zero
2. If water is flowing in a pipe with speed 5 m/s, then its kinetic energy per unit volume is
- (1) 12500 J/m³
- (2) 15200 J/m³
- (3) 2000 J/m³
- (4) 10000 J/m³
3. Consider the following statements out of which one is labelled as Assertion (A) and other as Reason (R).
Assertion (A): According to the law of equipartition of energy, internal energy of an ideal gas at given temperature, is equally distributed between rotational and translational kinetic energies.
Reason (R): Rotational kinetic energy of monoatomic gas is non-zero.
- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

4. The molar specific heats of an ideal gas at constant pressure and constant volume are denoted by C_P and C_V respectively. If $\gamma = \frac{C_P}{C_V}$ and R is the universal gas constant, then C_P is equal to

(1) $\frac{R}{\gamma-1}$

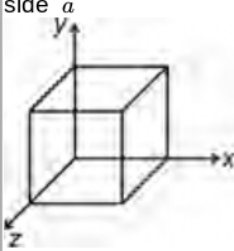
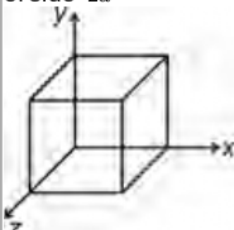
(2) $\frac{R\gamma}{\gamma-1}$

(3) $\frac{R\gamma}{1+\gamma}$

(4) $\frac{\gamma-1}{R}$

5. A Carnot cycle for a heat engine consists of
- (1) Two adiabatic and two isobaric processes
- (2) Two adiabatic and two isochoric processes
- (3) Two adiabatic and two isothermal processes
- (4) One adiabatic and three isothermal processes
6. A body cools in 4 minutes from 60°C to 50°C. The temperature of the body after the next 4 minutes will be (Assume the temperature of surrounding to be 10°C)
- (1) 42°C
- (2) 48°C
- (3) 35°C
- (4) 32°C

7. A variable electric field given as $\vec{E} = \alpha x \hat{i}$ where x represents x -coordinate, exists in a region. For the given information, match the columns and choose the correct option.

	Column-I		Column-II
(A)	Electric flux through the area vector $\vec{A} = a\hat{j}$ positioned at $y = a$	(P)	$8\alpha a^3$
(B)	Electric flux through the area vector $\vec{A} = a\hat{i}$ positioned at $x = a$	(Q)	$\alpha a^3 \epsilon_0$
(C)	Net charge inside the given cube of side a 	(R)	Zero
(D)	Net electric flux through the given cube of side $2a$ 	(S)	αa^2

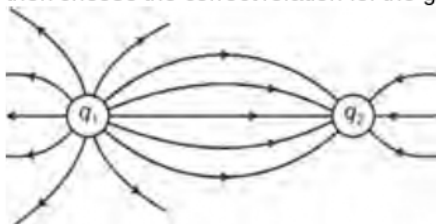
(1) (A)-R; (B)-S; (C)-P; (D)-Q

(2) (A)-R; (B)-S; (C)-Q; (D)-P

(3) (A)-P; (B)-Q; (C)-R; (D)-S

(4) (A)-P; (B)-R; (C)-Q; (D)-S

8. Electric field lines for two charges q_1 and q_2 is given below, then choose the correct relation for the given charges.



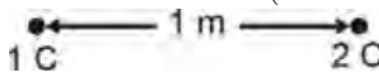
(1) $|q_1| = |q_2|$

(2) $|q_1| > |q_2|$

(3) $|q_1| < |q_2|$

(4) None of the above

9. Two-point charges are placed apart at 1 m distance. Minimum work done by external force to separate them to infinite distance will be (Assume $K = \frac{1}{4\pi\epsilon_0}$)



(1) $2K$

(2) $-2K$

(3) Zero

(4) $2.5K$

10. The number of electrons flowing per second in a resistor which is dissipating power of 110 W having potential difference of 220 V, is

(1) 1.25×10^{19}

(2) 31.25×10^{17}

(3) 6.25×10^{30}

(4) 1.75×10^9

11. Two conductors have the same resistance at 0°C and their temperature coefficients of resistance are α_1 and α_2 . The equivalent temperature coefficient of their series combination is

(1) $\alpha_1 + \alpha_2$

(2) $\alpha_1 \alpha_2$

(3) $\alpha_1 - \alpha_2$

(4) $\frac{\alpha_1 + \alpha_2}{2}$

12. The minimum capacitance that can be obtained by combining three identical capacitors of capacity $3 \mu\text{F}$ is

(1) $\frac{1}{2} \mu\text{F}$

(2) $6 \mu\text{F}$

(3) $1 \mu\text{F}$

(4) $9 \mu\text{F}$

13. A bar magnet of magnetic moment M is cut into two equal parts of equal length. Now they are placed on each other in such a way that similar poles are in same side. The new magnetic moment of the combination will be

(1) $\frac{M}{2}$

(2) M

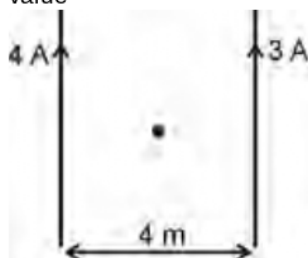
(3) $2M$

(4) Zero

14. A charge q moves with velocity $\vec{v}$ in a region where electric field $\vec{E}$ and magnetic field $\vec{B}$ both exist, then net force on the charge will be

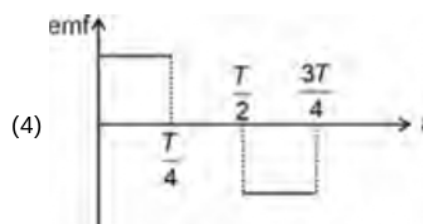
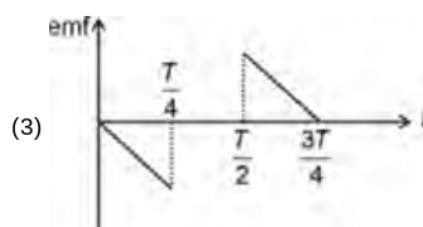
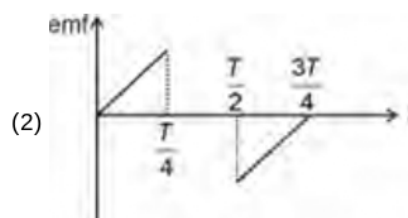
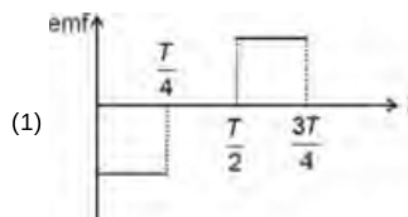
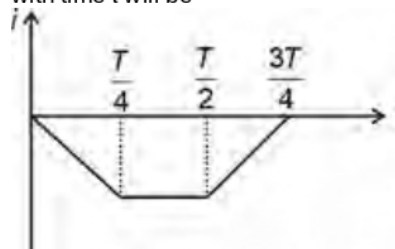
- (1) $q\vec{B} + q(\vec{E} \times \vec{v})$
- (2) $q\vec{B} + q(\vec{v} \times \vec{E})$
- (3) $q\vec{E} + q(\vec{v} \times \vec{B})$
- (4) $q\vec{E} + q(\vec{B} \times \vec{v})$

15. Two long parallel wires at a distance 4 m apart are carrying current 4.0 A and 3.0 A as shown in figure. The net magnetic field at mid-point between the wires will have value



- (1) $\frac{\mu_0}{2\pi}$
- (2) $\frac{\mu_0}{4\pi}$
- (3) $\frac{\mu_0}{\pi}$
- (4) $\frac{2\mu_0}{\pi}$

16. The current i in a coil having self inductance L varies with time t as shown in the figure. The variation of induced emf with time t will be



17. Which of the following devices work on the eddy current effect?

- (1) Electric heater
- (2) A.C. generator
- (3) A.C. motor
- (4) Induction furnace

18. A small signal voltage $V(t) = V_0 \sin \omega t$ is applied across an ideal inductor having inductance L . Then

- (1) Current $I(t)$ is in phase with voltage $V(t)$
- (2) Current $I(t)$ leads voltage $V(t)$ by 180° phase
- (3) Current $I(t)$ lags voltage $V(t)$ by 90° phase
- (4) Over a full cycle, the inductor dissipates some energy from the voltage source

19. In an A.C. circuit, the emf (E) and the current (I) at any instant are given respectively by $E = E_0 \sin \omega t$, $I = I_0 \sin(\omega t + \phi)$. If $0 < \phi < \frac{\pi}{2}$, then
- Circuit may contain resistor only
 - Circuit may contain inductor only
 - Circuit may contain resistor and inductor only
 - Circuit may contain resistor and capacitor only
20. Given below are two statements.
Statement (I): Nuclear forces are always attractive in nature.
Statement (II): Nuclear force does not depend on the electric charge of the nucleon.
 In the light of the above statements, choose the most appropriate option from the following.
- Statement (I) is correct but statement (II) is incorrect.
 - Statement (I) is incorrect but statement (II) is correct
 - Both statements (I) and (II) are correct
 - Both statements (I) and (II) are incorrect.
21. A hydrogen atom in its ground state absorbs 12.09 eV of energy. Its angular momentum is increased by (h is Planck's constant)
- $\frac{h}{2\pi}$
 - $\frac{h}{\pi}$
 - $\frac{2h}{\pi}$
 - $\frac{3h}{2\pi}$
22. When light falls on a photosensitive surface, electrons are emitted from the surface. The maximum kinetic energy of these electrons does not depend on
- the wavelength of incident light
 - the frequency of incident light
 - the intensity of the light
 - the type of material used as emitter plate
23. In a closed organ pipe speed of sound is 340 m s^{-1} and fundamental frequency is 340 Hz. If its length is 23.5 cm, then value of end correction is
- 0.75 cm
 - 1 cm
 - 1.25 cm
 - 1.5 cm
24. For detecting intensity of light, we use
- Photo-diode in forward bias
 - Photo-diode in reverse bias
 - Photo-diode in no biasing condition
 - LED in reverse bias
25. Carbon, Silicon and Germanium atoms have same lattice structure. Their valence and conduction bands are separated by energy band gaps represented by $(E_g)_C$, $(E_g)_{Si}$ and $(E_g)_{Ge}$ respectively. Which one of the following relationships is true?
- $(E_g)_C < (E_g)_{Ge}$
 - $(E_g)_{Ge} > (E_g)_{Si}$
 - $(E_g)_C > (E_g)_{Si}$
 - $(E_g)_C = (E_g)_{Ge}$
26. The extension produced in a wire by the application of load is 5 mm. The extension in a wire of the same material and length but half the radius, by the same load is
- 40 mm
 - 20 mm
 - 50 mm
 - 30 mm
27. The equation of a wave travelling on a string is given by $y = 4 \sin \frac{\pi}{2} \left(8t - \frac{x}{8} \right)$, the velocity of the wave is (where x and y are in cm and t is in seconds)
- 64 cm/s in $-x$ direction
 - 32 cm/s in $-x$ direction
 - 64 cm/s in $+x$ direction
 - 32 cm/s in $+x$ direction
28. When the displacement of a particle executing SHM (in horizontal plane) is half of its amplitude, the ratio of its kinetic energy to potential energy is
- 1 : 3
 - 2 : 1
 - 3 : 1
 - 1 : 2
29. The ratio of contribution by the electric field and magnetic field components to the intensity of an electromagnetic wave is (c = speed of EM wave)
- $c : 1$
 - $1 : c$
 - $1 : 1$
 - $1 : c^2$

30. In Young's double slit experiment, match the following two columns.

Column I	Column II
(a) When apparatus is immersed in a liquid	(i) Fringe width will increase
(b) When wavelength of light used is increased	(ii) Fringe width will decrease
(c) When monochromatic light is replaced by white light	(iii) Fringe width will remain constant
(d) When distance between slits (d) is doubled and distance from screen (D) is also doubled	(iv) Bright white fringe at centre

- (1) a(ii); b(i); c(iv); d(iii)
 (2) a(i); b(ii), c(iii), d(iv)
 (3) a(ii), b(iii), c(i), d(iv)
 (4) a(ii), b(i), c(iii), d(iv)

31. A metre stick is balanced on a fulcrum at its centre. When two stones each of mass 10 g are put on top of the other at 20 cm mark, the stick is found to be balanced at 49 cm. The mass of the metre stick is

- (1) 0.145 kg
 (2) 0.29 kg
 (3) 0.58 kg
 (4) 1.16 kg

32. A ring of mass 4 kg rotates about its diameter with angular speed 1000 rad/s. If ring has radius 0.25 m, then magnitude of angular momentum of ring about this axis is

- (1) 62.5 J s
 (2) 125 J s
 (3) 250 J s
 (4) 500 J s

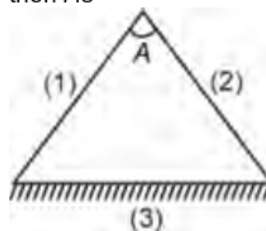
33. Two satellites of earth, S_1 and S_2 of same mass, are moving in the same orbit of radius r . Now only S_2 is taken to a new orbit of radius $4r$. Then which one of the following statements is true?

- (1) Ratio of their kinetic energy is 1 : 4
 (2) Ratio of their time period is 1 : 8
 (3) Ratio of their total energy is 1 : 8
 (4) Ratio of their angular momentum is 2 : 1

34. Work done by gravitational force in moving a particle of mass m from the surface of earth to a point at a depth $\frac{R}{2}$ from the surface is (where R is radius of earth).

- (1) $\frac{3}{8}mgR$
 (2) $\frac{mgR}{3}$
 (3) $-\frac{mgR}{4}$
 (4) $-\frac{mgR}{2}$

35. An equilateral prism has its bottom surface (3) silvered as shown below. Light rays falling at an angle of incidence i on surface (1) retrace their path after reflection from the silvered surface. If refractive index of prism material is $\sqrt{2}$, then i is



- (1) Zero
 (2) 30°
 (3) 45°
 (4) 60°

36. Given below are two statements:

Statement (I): For a simple microscope, the angular size of the object is less than the angular size of the image, when the lens is kept close to the eye.

Statement (II): Thick lenses give coloured images due to dispersion.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statements (I) and (II) are correct.
 (2) Both statements (I) and (II) are incorrect
 (3) Statement (I) is correct but (II) is incorrect
 (4) Statement (I) is incorrect but (II) is correct

37. The number of significant figures in 0.0305 is

- (1) Two
 (2) Three
 (3) Four
 (4) Six

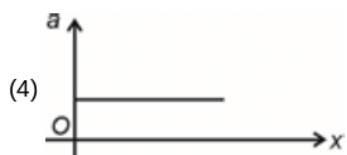
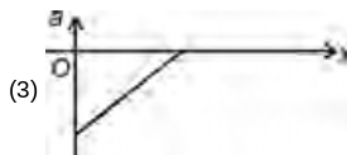
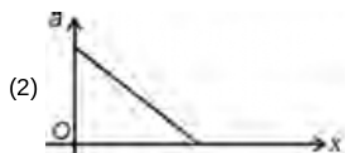
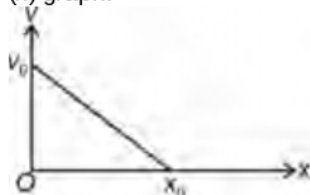
38. A new system of units is proposed in which unit of mass is γ kg, unit of length is β m and unit of time is α s. How much will 10 J measure in this new system?

- (1) $10\alpha\beta\gamma$
- (2) $10\frac{\alpha\beta}{\gamma}$
- (3) $10\alpha^2\gamma^{-1}\beta^{-2}$
- (4) $10\alpha^3\gamma\beta^{-2}$

39. A lift is coming from 10th to 5th floor with decreasing speed. If ground floor is taken as origin and upward direction as positive. Then choose the correct option regarding displacement (x), velocity (v) and acceleration (a).

- (1) $x < 0, v < 0, a > 0$
- (2) $x > 0, v < 0, a < 0$
- (3) $x < 0, v > 0, a < 0$
- (4) $x > 0, v > 0, a > 0$

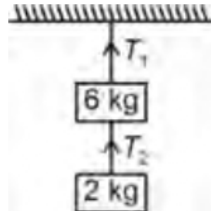
40. For the given velocity (v)-displacement (x) graph of a particle, choose the correct acceleration (a)-displacement (x) graph.



41. A body of mass 10 kg is acted upon by two perpendicular forces 3 N and 4 N. The resultant acceleration of the body is

- (1) 0.5 m/s^2 at an angle of 53° w.r.t. 4 N
- (2) 0.2 m/s^2 at an angle of 53° w.r.t. 4 N
- (3) 0.5 m/s^2 at an angle of 37° w.r.t. 4 N
- (4) 1 m/s^2 at an angle of 37° w.r.t. 3 N

42. Two masses of 6 kg and 2 kg are suspended with the help of massless inextensible strings as shown in figure. Calculate T_1 and T_2 when whole system is going upwards with an acceleration of 2 m/s^2 .



- (1) $T_1 = 96 \text{ N}, T_2 = 24 \text{ N}$
- (2) $T_1 = 24 \text{ N}, T_2 = 96 \text{ N}$
- (3) $T_1 = 16 \text{ N}, T_2 = 96 \text{ N}$
- (4) $T_1 = 36 \text{ N}, T_2 = 76 \text{ N}$

43. Given below are two statements one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Potential energy is positive, if conservative field is repulsive in nature.

Reason (R): Potential energy is negative, if conservative field is attractive in nature.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false

44. The potential energy for a conservative force system is given by $U = bx^2 - cx$, where b and c are constants. The equilibrium position for the system is
- (1) $\frac{2b}{c}$
 - (2) $\frac{3b}{c^2}$
 - (3) $\frac{c}{2b}$
 - (4) $2bc$
45. A particle is projected from ground at an angle θ with the horizontal such that maximum height reached by the particle is $\frac{1}{4}$ times of the horizontal range of the particle. Value of θ will be
- (1) 37°
 - (2) 53°
 - (3) 45°
 - (4) 76°

CHEMISTRY

46. Match the column I containing mixture/pairs of compounds with their respective best separation techniques given in column II

	Column I		Column II
a.	Chloroform and aniline	(i)	Sublimation
b.	Glycerol from spent-lye	(ii)	Distillation under reduced pressure
c.	Aniline – water mixture	(iii)	Distillation
d.	Calcium sulphate and camphor	(iv)	Steam distillation

Choose the correct match from the options given below:

- (1) a–(ii), b–(i), c–(iv), d–(iii)
- (2) a–(iii), b–(iv), c–(ii), d–(i)
- (3) a–(ii), b–(iii), c–(i), d–(iv)
- (4) a–(iii), b–(ii), c–(iv), d–(i)

47. The correct order of acidic strength of given carboxylic acids is

- (1) $(\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH} < \text{ClCH}_2\text{COOH} > \text{Cl}_2\text{CHCOOH}$
- (2) $(\text{CH}_3)_3\text{CCOOH} > \text{Cl}_2\text{CHCOOH} > (\text{CH}_3)_2\text{CHCOOH} > \text{ClCH}_2\text{COOH}$
- (3) $\text{Cl}_2\text{CHCOOH} > \text{ClCH}_2\text{COOH} > (\text{CH}_3)_3\text{CCOOH} > (\text{CH}_3)_2\text{CHCOOH}$
- (4) $\text{Cl}_2\text{CHCOOH} > \text{ClCH}_2\text{COOH} > (\text{CH}_3)_2\text{CHCOOH} > (\text{CH}_3)_3\text{CCOOH}$

48. Consider the following statements

Statement I: Isopentane and neopentane are chain isomers.

Statement II: Ethanoic acid and Methyl ethanoate are functional group isomers.

In the light of above statements, choose the correct option.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

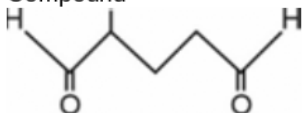
49. Choose the incorrect match

- (1) Water gas – $\text{CO} + \text{H}_2$
- (2) Producer gas – $\text{CO} + \text{CH}_4$
- (3) Cryolite – Na_3AlF_6
- (4) Borax – $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$

50. Total number of peroxide linkage(s) in CrO_5 is

- (1) 0
- (2) 1
- (3) 2
- (4) 3

51. Compound



is obtained on the reductive ozonolysis of

- (1)
- (2)
- (3)
- (4)

52. Given below are the two statements

Statement I: Bases present in DNA are adenine, guanine, cytosine and thymine.

Statement II: Adenine and cytosine are bicyclic compounds.

In light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

53. If the radius of the first Bohr orbit of He^+ is a Å, then the radius of the third Bohr orbit of hydrogen atom will be

- (1) 4a Å
- (2) 18a Å
- (3) 9a Å
- (4) 6a Å

54. Most acidic compound among the following is

- (1)
- (2)
- (3)
- (4)

55. Match list I with list II

	List I Name of compound		List II Structure
(a)	$\beta\text{-D-(+)-Glucopyranose}$	(i)	
(b)	$\alpha\text{-D-(-)-Fructofuranose}$	(ii)	
(c)	D-(+)-Glyceraldehyde	(iii)	
(d)	$\beta\text{-D-2-deoxyribose}$	(iv)	

The correct match is

- (1) (a)–(iii), (b)–(i), (c)–(ii), (d)–(iv)
- (2) (a)–(ii), (b)–(i), (c)–(iv), (d)–(iii)
- (3) (a)–(i), (b)–(iv), (c)–(ii), (d)–(iii)
- (4) (a)–(iv), (b)–(i), (c)–(ii), (d)–(iii)

56. Given below are the two statements.

Statement I: In photoelectric effect, the number of electrons ejected is proportional to the frequency of the light used.

Statement II: Photoelectric effect can be explained by the particle nature of the electromagnetic radiation.

In light of above statements, choose the correct answer.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II incorrect
- (4) Statement I is incorrect but statement II is correct

57. Given below are two statements.

Statement I : The magnitude of hydrogen bonding depends on the physical state of the compound.

Statement II: Hydrogen bonding is maximum in the gaseous state and minimum in the solid state

In the light of above statements, choose the correct answer.

- (1) Statement I is correct but statement II is incorrect
- (2) Statement I is incorrect but statement II is correct
- (3) Both statement I and statement II are correct
- (4) Both statement I and statement II are incorrect

58. Consider the following statements

(a) AgCl has more ionic character as compared to NaCl

(b) Both SO_2 and H_2O have bent shape

(c) Dipole moment of HI is less than that of HCl

The correct statements are.

- (1) a and b only
- (2) a and c only
- (3) b and c only
- (4) a, b and c

59. If the observed dipole moment of A–B molecule is 1 D and A – B bond length is 100 pm then percentage ionic character in A–B molecule will be

- (1) 25.75%
- (2) 16.65%
- (3) 20.81%
- (4) 14.29%

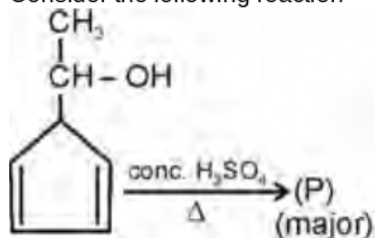
60. For $[\text{CrCl}_3 \cdot x\text{NH}_3]$, elevation in boiling point of one molal aqueous solution is equal to that of one molal aqueous solution of glucose, the value of x is

- (1) 3
- (2) 4
- (3) 5
- (4) 6

61. If 0.1 molal monobasic acid is 10% ionised in an aqueous solution, then the freezing point of the solution will be (given $K_f = 1.86 \text{ K m}^{-1}$)

- (1) -0.205°C
- (2) -0.102°C
- (3) -0.410°C
- (4) -0.302°C

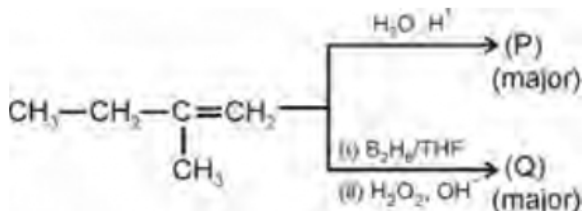
62. Consider the following reaction



The major product 'P' is

- (1)
- (2)
- (3)
- (4)

63.



Product (P) and (Q) can be best distinguished by

- (1) Sodium metal
- (2) Lucas test
- (3) Esterification reaction
- (4) Neutral FeCl_3

64. Given below are two statements
Statement I: The heating of phenyl ethyl ether with HI results in the formation of phenol and ethyl iodide.
Statement II: Butan-1-ol has higher boiling point than Ethoxyethane.
 In the light of above statements, choose the most appropriate answer from the options given below.
- Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
 - Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
65. Consider the following statements:
Statement I: Hot and concentrated sulphuric acid oxidises copper and evolves sulphur trioxide gas.
Statement II: Charring action of sulphuric acid shows its dehydrating nature.
 In the light of above statements, choose the most appropriate answer from the options given below.
- Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
 - Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
66. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R).
Assertion (A): PCl_3 fumes in moisture.
Reason (R): PCl_3 hydrolyses in the presence of moisture to evolve HCl gas.
 In the light of above statements, choose the most appropriate answer from the options given below:
- Both (A) and (R) are true and (R) is the correct explanation of (A)
 - Both (A) and (R) are true but (R) is not the correct explanation of (A)
 - Both (A) and (R) are false
 - (A) is true but (R) is false
67. Given below are two statements
Statement I: Fluorine is a stronger oxidising agent than chlorine in aqueous medium.
Statement II: F_2 has lower bond dissociation enthalpy than Cl_2 .
 In the light of above statements, choose the correct option.
- Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
 - Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
68. The weight of silver which is displaced by same quantity of electricity which displaces 2800 mL of chlorine gas at STP will be
- 13.5 g
 - 27 g
 - 54 g
 - 108 g
69. Which of the following can act as both Bronsted acid as well as Bronsted base?
- HPO_4^{2-}
 - H_3PO_3
 - HPO_3^{2-}
 - PO_4^{3-}
70. Limiting molar conductivity of CH_3COOH (i.e. $\Lambda_m^\circ(\text{CH}_3\text{COOH})$) is equal to
- $\Lambda_m^\circ(\text{NaCl}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{CH}_3\text{COOH})$
 - $\Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl})$
 - $\Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{NaOH})$
 - $\Lambda_m^\circ(\text{NaOH}) + \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl})$
71. If 1 L of 0.2 M HCl, 2L of 0.1 M HCl and 1 L of 0.2 M H_2SO_4 are mixed together, then the pH of the resulting mixture will approximately be
- 0.2
 - 0.5
 - 0.7
 - 0.8
72. Given below are two statements,
Statement I: The maximum covalency of boron is 3.
Statement II: The covalency of Al in $[\text{AlCl}(\text{H}_2\text{O})_5]^{2+}$ is 6.
 In light of above statements, choose the correct option.
- Both statement I and statement II are correct
 - Both statement I and statement II are incorrect
 - Statement I is correct but statement II is incorrect
 - Statement I is incorrect but statement II is correct
73. For the given ions, the correct order of ionic size is
- $\text{K}^+ < \text{Ca}^{2+} < \text{S}^{2-}$
 - $\text{S}^{2-} < \text{Ca}^{2+} < \text{K}^+$
 - $\text{Ca}^{2+} < \text{K}^+ < \text{S}^{2-}$
 - $\text{S}^{2-} < \text{K}^+ < \text{Ca}^{2+}$

74. Match list-I with list-II

List-I Property	List-II (Inner transition element)
(a) The lanthanoid which exhibits +4 oxidation state but only in oxide (MO_2)	(i) Np
(b) The actinoid which exhibits $[\text{Rn}]5f^7 6d^1 7s^2$ ground state electronic configuration	(ii) Pr
(c) The actinoid which exhibits +7 oxidation state	(iii) Sm
(d) The lanthanoid with $[\text{Xe}] 4f^5$ electronic configuration in +3 oxidation state	(iv) Cm

Choose the correct option.

- (1) (a)–(ii), (b)–(iv), (c)–(i), (d)–(iii)
 (2) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)
 (3) (a)–(i), (b)–(ii), (c)–(iii), (d)–(iv)
 (4) (a)–(iv), (b)–(ii), (c)–(iii), (d)–(i)

75. The colour of chromate ion in aqueous solution is

- (1) Green
 (2) Yellow
 (3) Purple
 (4) White

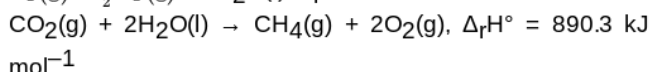
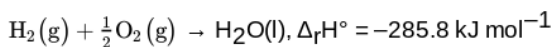
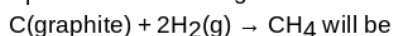
76. Given below are two statements,

Statement I: The molar enthalpy of formation for both C (graphite) and S(s), rhombic at 298 K is zero.

Statement II: According to third law of thermodynamics, the entropy of any pure crystalline substance approaches zero as the temperature approaches absolute zero.

In light of above statements, choose the correct option.

- (1) Both statement I and statement II are correct
 (2) Both statement I and statement II are incorrect
 (3) Statement I is correct but statement II is incorrect
 (4) Statement I is incorrect but statement II is correct

 77. $\text{C}(\text{graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}), \Delta_r H^\circ = -393.5 \text{ kJ mol}^{-1}$

 Based on the above thermochemical equation, the value of $\Delta_r H^\circ$ at 298 K for the given reaction will be


- (1) $144.0 \text{ kJ mol}^{-1}$
 (2) $-144.0 \text{ kJ mol}^{-1}$
 (3) $-74.8 \text{ kJ mol}^{-1}$
 (4) $-64.7 \text{ kJ mol}^{-1}$

78. Weight of lime (CaO) obtained on decomposition of 200 g of 95% pure limestone is

- (1) 112g
 (2) 106.4 g
 (3) 160.4 g
 (4) 112.8 g

79. Match list I with list II

List I Species	List II Number of unpaired electron(s) on central atom
(a) $[\text{Cr}(\text{NH}_3)_6]^{3+}$	(i) 4
(b) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$	(ii) 0
(c) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$	(iii) 3
(d) $[\text{MnCl}_6]^{3-}$	(iv) 2

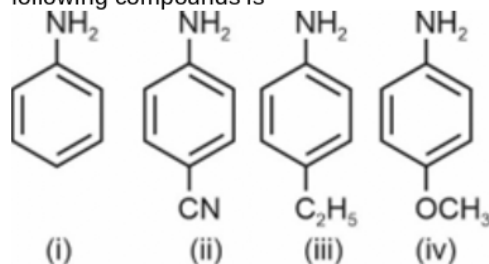
The correct match is

- (1) (a)–(iii), (b)–(ii), (c)–(iv), (d)–(i)
 (2) (a)–(ii), (b)–(iii), (c)–(i), (d)–(iv)
 (3) (a)–(i), (b)–(iii), (c)–(ii), (d)–(iv)
 (4) (a)–(i), (b)–(ii), (c)–(iv), (d)–(iii)

80. Heteroleptic complex among the following is

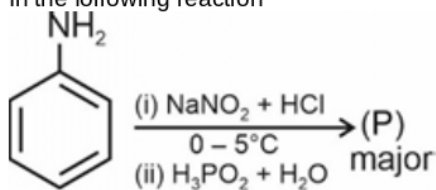
- (1) $\text{K}_2[\text{Zn}(\text{OH})_4]$
 (2) $\text{Hg}[\text{Co}(\text{SCN})_4]$
 (3) $[\text{Pt}(\text{NH}_3)_2\text{Cl}(\text{NO}_2)]$
 (4) $\text{K}_3[\text{Fe}(\text{CN})_6]$

81. The correct increasing order of basic strength of the following compounds is



- (1) (iii) > (iv) > (i) > (ii)
 (2) (iv) > (iii) > (i) > (ii)
 (3) (i) > (ii) > (iii) > (iv)
 (4) (ii) > (iii) > (iv) > (i)

82. In the following reaction



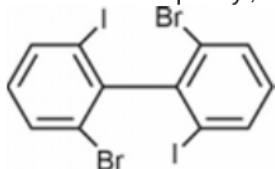
The product (P) will be

- (1) Oc1ccccc1
- (2) [O-][N+](=O)c1ccccc1
- (3) Clc1ccccc1
- (4) c1ccccc1

83. Given below are two statements

Statement I: The total possible monochloro isomers expected to be formed on free radical monochlorination of 2-Methylbutane is six.

Statement II: Biphenyl,

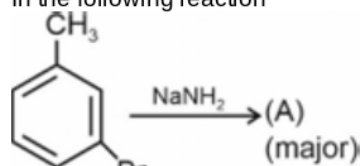


is optically active

In the light of above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are correct
- (2) Both statement I and statement II are incorrect
- (3) Statement I is correct but statement II is incorrect
- (4) Statement I is incorrect but statement II is correct

84. In the following reaction



The product (A) will be

- (1) Nc1cccc(C)c1
- (2) Nc1ccccc1C
- (3) Cc1ccccc1
- (4) BrC1=CC=CC=C1C

85. Half life of a zero order reaction is 15 min. The time required to consume the reactant completely is

- (1) 20 min
- (2) 4 min
- (3) 30 min
- (4) 60 min

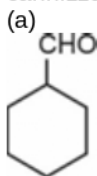
86. If rate of reaction gets doubled on increase of every 10 °C rise of temperature then by what factor rate of reaction will change if temperature is increased from 30 °C to 60 °C?

- (1) Increased by 6 times
- (2) Decreased by 6 times
- (3) Increased by 8 times
- (4) Decreased by 8 times

87. Correct reactivity order of given compounds towards nucleophilic addition reaction is

- (1) CH3CHO > HCHO > C6H5CHO
- (2) C6H5CHO > CH3CHO > HCHO
- (3) HCHO > C6H5CHO > CH3CHO
- (4) HCHO > CH3CHO > C6H5CHO

88. Which among the following compound(s) will give cannizzaro reaction?



- (b) HCHO
(c) CH_3COCH_3
(d) CH_3CHO

- (1) (b) only
(2) (a) and (b) only
(3) (c) and (d) only
(4) (a), (b) and (d) only

89. Which among the following anions can be detected by the colour obtained on addition of $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$ to the aqueous salt solution?

- (1) Cl^-
(2) Br^-
(3) SO_4^{2-}
(4) S^{2-}

BIOLOGY

91. 'A' occupies the distant most region of stem axis.
In the light of above statement, choose the option which **correctly** fills in the blank for 'A'.

- (1) Lateral meristem
(2) Shoot apical meristem
(3) Intercalary meristem
(4) Root apical meristem

92. Which of the given nitrogenous bases are having heterocyclic, 6-membered single-ring structure?

- a. Adenine
b. Thymine
c. Guanine
d. Cytosine
e. Uracil

The **correct** ones are

- (1) c and d only
(2) a and b only
(3) b, d and e only
(4) a and c only

90. Given below are two statements

Statement I: Ceric ammonium nitrate gives a red colouration when reacts with ethanol.

Statement II: Ethanol gives positive iodoform test.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (1) Both statement I and statement II are correct
(2) Both statement I and statement II are incorrect
(3) Statement I is correct but statement II is incorrect
(4) Statement I is incorrect but statement II is correct

93. Read the given Assertion (A) and Reason (R) and choose the correct option.

Assertion (A): The two sites in the large subunit of ribosome are close enough and the formation of peptide bond between subsequent amino acid would be favoured energetically.

Reason (R): The presence of ribozyme would enhance the rate of peptide bond formation.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) Both (A) and (R) are false
(4) (A) is true but (R) is false

94. Match column I with column II

	Column I		Column II
a.	Dicot stem	(i)	It consists of large, empty and colourless cells which help in rolling of leaves during drought
b.	Dicot root	(ii)	Shows the presence of scattered vascular bundles
c.	Monocot stem	(iii)	Presence of almost uniform sized vascular bundles and collenchymatous hypodermis
d.	Monocot leaf	(iv)	Presence of tetrarch vascular bundles epidermis is devoid of cuticle

Choose the **correct** option

- (1) a(iv), b(i), c(iii), d(ii)
- (2) a(iii), b(iv), c(ii), d(i)
- (3) a(i), b(ii), c(iii), d(iv)
- (4) a(iii), b(iv), c(i), d(ii)

95. The actual structure of tRNA

- (1) Looks like a clover-leaf
- (2) Is not a three-dimensional structure
- (3) Looks like inverted L-shape
- (4) Was given by Francis Crick and Watson

96. The only dead element of phloem tissue

- a. Have pointed needle-like apices
- b. Are found in secondary phloem
- c. Are used commercially for various purposes
- d. Have branches
- e. Are helpful in maintaining pressure gradient in sieve tubes.

The **correct** ones are

- (1) b, c and d
- (2) a and e
- (3) a, b and c
- (4) a, b and d

97. Which among the following enzymes facilitates opening of DNA double helix during transcription?

- (1) DNA ligase
- (2) DNA helicase
- (3) RNA polymerase
- (4) DNA polymerase

98. Select the **mismatched** pair

(1)	Microsatellites	–	1-6 bp
(2)	Dystrophin	–	Largest known human gene
(3)	SSBP	–	Stabilises the exposed single strands of DNA during unwinding of helix
(4)	Introns	–	Required for efficient translation process

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

99. Which of the following chromosomes is also known as isobrachial chromosome?

- (1) Metacentric chromosome
- (2) Sub-metacentric chromosome
- (3) Acrocentric chromosome
- (4) Telocentric chromosome

100. How many chromatids will be there in the pachytene stage of a cell, which has 15 tetrads?

- (1) 60
- (2) 30
- (3) 15
- (4) 100

101. State the following statements as true (T) or false (F) and choose the **correct** option.

- (a) Neutral solutes may move across the membrane by the process of simple diffusion along the concentration gradient.
- (b) The peripheral proteins lie on the surface of the membrane.
- (c) According to fluid mosaic model given by Nicolson and Singer, the quasi fluid nature of protein enables lateral movement of lipids.
- (d) Na^+/K^+ pump requires ATP to transport molecules across membrane.

	(a)	(b)	(c)	(d)
(1)	T	T	F	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	F	T	F	F

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

102. Duplication of semiautonomous organelles is associated with which phase of cell cycle?

- (1) G₂ phase
- (2) G₁ phase
- (3) S phase
- (4) M phase

103. Which among the following events take place during meiosis I but not during mitosis?

- a. Nuclear membrane disappears
- b. Pairing of homologous chromosomes
- c. Terminalization of chiasmata
- d. Microtubules attach to kinetochore of chromosomes
- e. Separation of homologous chromosome
- f. Arrangement of chromosomes on equatorial plate

Choose the **correct** ones

- (1) a, b and e only
- (2) b and d only
- (3) b, c and e only
- (4) c, d, e and f only

104. Bryophytes include various mosses and liverworts that are found commonly growing in moist shaded areas in the hills. In this group

- (1) Zygotes undergo reduction division immediately after formation.
- (2) Sexual reproduction is oogamous type involving motile male and non-motile female gametes.
- (3) Sporophytes are free-living and photosynthetic.
- (4) Main plant body produces gametes by meiosis

105. Select the **incorrect** features w.r.t. brown algae.

- a. Gametes are pyriform and bear two lateral flagella
- b. Presence of chlorophyll a and c
- c. Cell wall has polysulphate esters
- d. They vary in colour from brown to olive green depending upon the amount of the xanthophyll called fucoxanthin.
- e. Stored food is similar to amylopectin and glycogen in structure.

- (1) a, c and d
- (2) Only d
- (3) c and e
- (4) a, b and c

106. Choose the odd one w.r.t components of a nucleotide.

- (1) Monosaccharide
- (2) Phosphoric acid
- (3) Heterocyclic nitrogenous base
- (4) Homocyclic moiety

107. In which year, the two enzymes responsible for restricting the growth of bacteriophage in *Escherichia coli* were isolated?

- (1) 1962
- (2) 1965
- (3) 1967
- (4) 1963

108. Consider the following features:

- (a) Enables glucose transport into cells
- (b) Heteropolymeric structure
- (c) Belongs to the category of biomacromolecule that is similar to antibodies

Select the option for which all the above given features hold true.

- (1) Trypsin
- (2) GLUT-4
- (3) Collagen
- (4) Carbonic anhydrase

109. Select the secondary metabolite among the following which can be categorised as both drug as well as an alkaloid.

- (1) Morphine
- (2) Abrin
- (3) Ricin
- (4) Concanavalin A

110. The type of muscle fibres present in biceps are similar to the type of muscle fibres present in heart as both have/show

- (1) Branched appearance
- (2) Uninucleated condition
- (3) Presence of striations
- (4) Intercalated discs

111. The problem of night blindness due to vitamin 'A' deficiency can be prevented by consuming

- (1) Golden rice
- (2) Bt corn
- (3) α -1-antitrypsin
- (4) Humulin

112. The non-excitable cells of the neural tissue have the/are

- (1) Ability to shorten in response to stimulation
- (2) Structures such as dendrites and cell body
- (3) Ability to divide
- (4) The structural and functional unit of neural tissue

113. Match the column I with column II w.r.t. an adult human.

Column I		Column II	
(a)	Clavicle	(i)	Atlas
(b)	Sternum	(ii)	Axis
(c)	1 st cervical vertebra	(iii)	Breastbone
(d)	2 nd cervical vertebra	(iv)	Collar bone

Choose the correct option.

- (1) (a)(iv), (b)(iii), (c)(ii), (d)(i)
- (2) (a)(iv), (b)(ii), (c)(iii), (d)(i)
- (3) (a)(i), (b)(ii), (c)(iii), (d)(iv)
- (4) (a)(iv), (b)(iii), (c)(i), (d)(ii)

114. A person that has B –ve blood group is compatible to receive blood from another person having

- (1) Anti-A antibodies in blood plasma and absence of 'Rh' antigens on the surface of RBCs
- (2) Anti-B antibodies in the blood plasma and presence of 'Rh' antigens on the surface of RBCs
- (3) Antigen 'A' and 'B' on the surface of RBCs and absence of 'Rh' antigens
- (4) All blood group antigens on the surface of RBCs including 'Rh' factor

115. Which of the following conditions decreases the cardiac output in humans?

- (1) Release of epinephrine from adrenal medulla
- (2) Neural signals through the sympathetic nerve fibres
- (3) Neural signals through the parasympathetic nerve fibres
- (4) Release of noradrenaline from the adrenal medulla

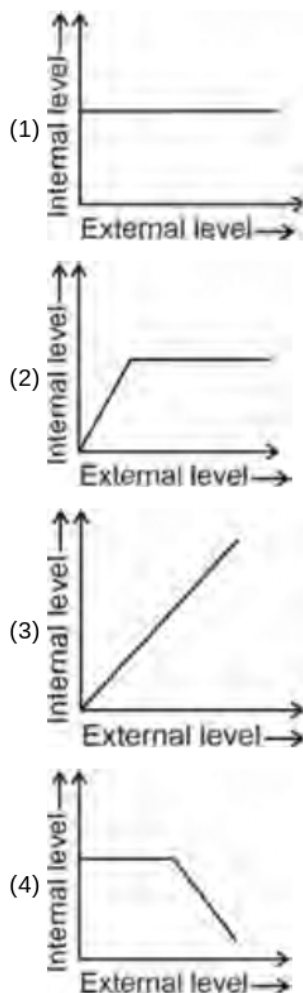
116. Read the following statements and select the **correct** option

Statement A: For more than hundred years, microbes are being used to treat sewage by the process of activated sludge formation.

Statement B: Microbes are used to ferment fish, soyabean and bamboo shoots to make foods.

- (1) Both statements are correct
- (2) Both statements are incorrect
- (3) Only statement A is correct
- (4) Only statement B is correct

117. Which graph **correctly** describes the response to abiotic factors of an overwhelming majority of animals.



118. All of the following options are related to *In-situ* conservation strategy of biodiversity, **except**

- (1) They are identified for maximum protection of certain regions with high degree of endemism.
- (2) Emphasizes on the protection of whole ecosystem
- (3) It includes protective maintenance of threatened species out from their natural habitat.
- (4) It includes Biosphere reserves, National Park and Wildlife sanctuaries

119. The extinction of passenger pigeon was due to

- (1) Non-availability of food
- (2) Over exploitation by humans
- (3) Increased number of predatory birds
- (4) Alien species invasion

120. Gametes are non-flagellated and similar in size in the sexual life cycle of

- (1) *Chlamydomonas*
- (2) *Fucus*
- (3) *Spirogyra*
- (4) *Polytrichum*

121. In anaerobic sludge digester, during digestion, bacteria produce a mixture of gases such as

- (1) CH₄, H₂S and CO₂
- (2) CH₄ and O₂ only
- (3) H₂S and CO₂ only
- (4) CH₄ and CO₂ only

122. In which of the following interactions, both partners are benefitted?

- (1) Predation
- (2) Competition
- (3) Mutualism
- (4) Parasitism

123. Match Column I with Column II and choose the correct option

	Column I		Column II
a.	Joseph Priestley	(i)	His experiments involved both aerobic bacteria and eukaryotic organisms like <i>Cladophora</i>
b.	Cornelius van Neil	(ii)	He performed series of experiments that revealed essential role of air in the growth of green plants
c.	T.W. Engelmann	(iii)	He showed that it is only green parts of the plant that could release oxygen
d.	Jan Ingenhousz	(iv)	He inferred that oxygen evolved by the green plant comes from H ₂ O not from CO ₂

- (1) a(i), b(ii), c(iv), d(iii)
- (2) a(iii), b(iv), c(i), d(ii)
- (3) a(ii), b(i), c(iv), d(iii)
- (4) a(ii), b(iv), c(i), d(iii)

124. Which among the following is **not** a plant factor that affects the rate of photosynthesis?

- (1) Orientation of leaves
- (2) Availability of sunlight
- (3) Amount of chlorophyll
- (4) Internal CO₂ concentration

125. Read the following statements and select the **correct** option.

Statement A: Final electron acceptor in electron transport chain receives electron from cytochrome c oxidase.

Statement B: During complete oxidation of a glucose molecule, only two steps of substrate level phosphorylation reaction occur.

- (1) Only statement A is correct
- (2) Only statement B is correct
- (3) Both statements A and B are correct
- (4) Both statements A and B are incorrect

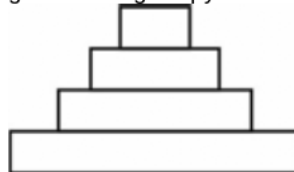
126. Which of the given organisms do not occupy the first trophic level or bottom trophic level in a food chain or pyramid?

- (1) Photoplanktons
- (2) Trees
- (3) Grasses
- (4) Zooplanktons

127. The six carbon containing intermediate formed during tricarboxylic acid cycle is

- (1) α ketoglutaric acid
- (2) Succinic acid
- (3) Citric acid
- (4) Malic acid

128. Which among the following statements are **true** for the given ecological pyramid?



- a. The food chain representing number of tree → herbivorous bird → Hawk can form such pyramid.
- b. It represents pyramid of number of forest ecosystem
- c. The length of each bar represents the quantity of parameter taken in consideration
- d. In such pyramid, the number of producer is always higher than the number of top carnivore
- e. In all types of ecosystems, this can represent the pyramid of energy flow

The **correct** ones are

- (1) a and b only
- (2) b, c and d only
- (3) d and e only
- (4) c, d and e only

129. Which of the following statements is **not** correct?

- (1) Isolated metabolic reactions *in-vitro* are not living things but surely living reactions.
- (2) Photoperiod affects reproduction in seasonal breeders, both plants and animals
- (3) Properties of tissues are not present in the constituent cells but arise as a result of interaction among the constituent cells
- (4) Taxonomy takes into account evolutionary relationships between organisms

130. Which among the following belongs to the class of fungi, commonly known as sac fungi?

- (1) *Rhizopus*
- (2) *Claviceps*
- (3) *Albugo*
- (4) *Ustilago*

131. Read the statements given below:

Statement A: One of the basic steps in genetically modifying an organism is maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

Statement B: Techniques to alter the chemistry of genetic material to suitably introduce them into respective host organism comes under the umbrella of genetic engineering. Choose the correct option.

- (1) Both statements A and B are incorrect
- (2) Both statements A and B are correct
- (3) Only statement A is correct
- (4) Only statement B is correct

132. Which of the following options is true for coronary artery?

- (1) Supplies oxygenated blood to the heart muscle
- (2) Supplies deoxygenated blood to lungs
- (3) Supplies oxygenated blood to brain
- (4) Supplies deoxygenated blood to uterine muscle.

133. Match column I with column II w.r.t. different types of neurons and their respective location/presence in humans.

Column I		Column II	
(a)	Multipolar neuron	(i)	Retina
(b)	Bipolar neuron	(ii)	Embryonic stage
(c)	Unipolar neuron	(iii)	Cerebral cortex

Select the correct option.

- (1) (a)(i), (b)(ii), (c)(iii)
- (2) (a)(iii), (b)(i), (c)(ii)
- (3) (a)(iii), (b)(ii), (c)(i)
- (4) (a)(ii), (b)(i), (c)(iii)

134. Which of the following sets of bones are equal in number in an adult human?

- (1) Wrist bone in one forelimb and cranial bones
- (2) Cranial bones and facial bones
- (3) Tarsals of one hindlimb and carpals of one forelimb
- (4) False ribs and cervical vertebrae

135.Assertion (A): Inulin unlike insulin is a heteropolysaccharide.

Reason (R): Inulin is composed of repeating units of more than one type of monosaccharides.

In the light of above statements, select the correct option.

- (1) Both (A and (R) are true and (R) is the correct explanation of (A).
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
- (3) (A) is true, (R) is false
- (4) Both (A) and (R) are false

136. While performing agarose gel electrophoresis, if a student fails to add ethidium bromide to the agarose gel, then

- (1) DNA fragments will still be visible under normal light
- (2) Bright orange coloured DNA bands will be visible under ionizing radiations
- (3) Elution will be easier to perform
- (4) DNA fragments will not be visible even after exposure to non-ionizing radiations

137. Select the **incorrect** option w.r.t PCR.

- (1) Can be employed for diagnosis of cancer
- (2) During the denaturation step, breaking of hydrogen bonds occur
- (3) Utilises dUTP, dCTP, ATP for extending the primers in the 5' → 3' direction
- (4) Two types of primers are required during the annealing step

138. In humans, repolarisation of neuronal membrane occurs due to rapid

- (1) Efflux of K^+
- (2) Influx of Na^+
- (3) Efflux of Na^+
- (4) Influx of K^+

139. Comprehend the statements given below:

- (a) All transformants are recombinants
 - (b) All recombinants are transformants
 - (c) All plasmids carry antibiotic resistance genes
 - (d) All selectable markers produce colour in the presence of a chromogenic substrate
 - (e) Restriction enzymes are found in prokaryotes
- Choose the option with only correct set.

- (1) (a), (b) and (e)
- (2) (c) and (e)
- (3) (a), (b), (c) and (d)
- (4) (b) and (e)

140. Mary Mallon continued to spread which of the following infectious diseases for several years through the food she prepared?

- (1) Typhoid
- (2) Pneumonia
- (3) Diphtheria
- (4) Smallpox

141. Select the function not performed by the hormones of fight or flight.

- (1) Glycogenolysis
- (2) Lipolysis and proteolysis
- (3) Increased respiratory rate
- (4) Pupillary constriction

142. Which of the following statements is **not true** about B-lymphocytes?

- (1) They produce army of immunoglobulins in response to a pathogen
- (2) They mediate humoral immune response
- (3) Their site of production is bone marrow
- (4) They primarily mediate graft rejection

143. "Although the neural system provides point-to-point coordination among organs, still regulation by endocrine system is required in human body". Select the most possible explanation for the above given statement.

- (1) Slow and short-lived response by former
- (2) Widespread response of the former
- (3) Ability of the former to continuously regulate cellular functions
- (4) Nerve fibres of former do not innervate all cells of the body

144. Which of the following is an incorrect statement regarding HIV infection?

- (1) Anti-retroviral drugs can completely cure HIV infection
- (2) The virus enters into macrophages where RNA genome of the virus replicates to form viral DNA with the help of enzyme reverse transcriptase.
- (3) There is always a time lag between the infection and appearance of AIDS symptoms
- (4) This infection does not spread by mere touch

145. Match column I with column II and select the correct option w.r.t classes of enzymes.

Column I		Column II	
(a)	Dehydrogenases	(i)	Class I
(b)	Ligases	(ii)	Class II
(c)	Lyases	(iii)	Class IV
(d)	Transferases	(iv)	Class VI

- (1) (a)(i), (b)(iv), (c)(iii), (d)(ii)
- (2) (a)(i), (b)(ii), (c)(iii), (d)(iv)
- (3) (a)(iv), (b)(iii), (c)(ii), (d)(i)
- (4) (a)(iii), (b)(ii), (c)(iv), (d)(i)

146. Choose the non-medicated IUD.

- (1) Multiload-375
- (2) Progestasert
- (3) LNG-20
- (4) Lippes loop

147. Given below are few features:

- (i) Body is circular in cross-section
 - (ii) Three germ layers are present but body cavity is not completely lined by mesoderm
 - (iii) Lacks digestive system thus, absorbs nutrients from the host directly through body surface
 - (iv) Have similar level of body organisation as *Nereis*
- How many of the above given features is/are correct for *Ancylostoma*?

- (1) Four
- (2) Three
- (3) Two
- (4) One

148. A researcher while sea diving found an organism 'X' with spiny body in ocean. On careful analysis in a lab, he found the presence of specialized water vascular system in the 'X' which helps in the locomotion, feeding and respiration. Identify 'X' and choose the correct option w.r.t. the features present in the members of the same phylum to which 'X' belongs to

- (1) They exhibit a well-developed excretory system
- (2) They are found in the marine water as well as in fresh water
- (3) Development is indirect with free-swimming larvae
- (4) Malpighian tubules are specialized excretory structures present in them

149. Assertion (A): Main challenge for production of insulin using rDNA technology was getting insulin assembled in its immature form.

Reason (R): Mature human insulin constitutes three peptide chains A, B and C attached to each other through peptide bonds.

In the light of above statements, choose the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) Both (A) and (R) are false
- (4) (A) is true but (R) is false

150. Fossils are generally found in

- (1) Igneous rocks
- (2) Sedimentary rocks
- (3) Metamorphic rocks
- (4) Both igneous and metamorphic rocks

151. A hominid, whose fossil was discovered in Java in 1891, had the cranial capacity around

- (1) 1650 cc
- (2) 400 cc
- (3) 900 cc
- (4) 1400 cc

152. Listed below are some features:

- (a) Separate sexes
- (b) Oviparity
- (c) Four-chambered heart
- (d) Internal fertilisation

How many of the above features is/are common for *Testudo* and *Crocodylus*?

- (1) One
- (2) Two
- (3) Three
- (4) Four

153. Nearly all of the essential nutrients and 70-80 per cent of electrolytes and water is reabsorbed in the part of renal tubule which

- (1) Performs only conditional reabsorption of Na^+
- (2) Also reabsorbs H^+ and K^+ selectively
- (3) Passively reabsorbs glucose
- (4) Is lined by simple cuboidal brush-border epithelium

154. Listed below are some factors

- (a) Gene flow
- (b) Genetic drift
- (c) No mutation
- (d) Random mating
- (e) Natural selection
- (f) Genetic recombination

Which of the above factors do not affect Hardy-Weinberg equilibrium?

- (1) a, b and d
- (2) c, d and e
- (3) c and d
- (4) b and e

155. Choose the **incorrect** statement among the following.

- (1) About 350 mya, fish with stout and strong fins could move on land and go back into water.
- (2) Sea weeds and few plants existed probably 320 mya.
The process of how non-cellular aggregates of giant macromolecule evolved into eukaryotic cells is now completely known.
- (3) In due course of time, single-celled chemoautotrophic organisms slowly evolved into multicellular life forms.
- (4) In due course of time, single-celled chemoautotrophic organisms slowly evolved into multicellular life forms.

156. Potato spindle tuber disease is caused by an infectious agent which

- (1) Has protein coat
- (2) Does not have any nitrogenous compound
- (3) Is similar in size to viruses
- (4) Has ribonucleic acid as its genetic material

157. Which of the following statement is **not** correct w.r.t. racemose inflorescence

- (1) Shoot axis continues to grow indefinitely
- (2) Younger flowers are present towards the apex and older flower are present at the base
- (3) Flowers are borne in basipetal order
- (4) It is seen in radish, mustard and lupin

158.Read the following statements (a-d)

- Flowers are bisexual with radial symmetry
- Bicarpellary, syncarpous, superior ovary with oblique septa and axile placentation
- Petal five, gamopetalous with valvate aestivation
- Stamen five, epipetalous and ditheous anthers.

The above given features are found in the members of family

- Malvaceae
- Fabaceae
- Solanaceae
- Poaceae

159.Identify the **mismatched** pair

(1)	Staminode	–	Sterile stamens
(2)	Diadelphous	–	Stamens are fused in two bundles
(3)	Polyandrous	–	Stamens remains free
(4)	Epiphyllous	–	Stamens are attached to petals

- (1)
- (2)
- (3)
- (4)

160.Identify the following statements as **true(T)** or **false(F)** and choose the option accordingly.

- Viruses cause diseases like mumps, herpes, influenza, AIDS and typhoid
- W.M. Stanley showed that viruses could be crystallised and crystals consist largely of protein.
- In general, the virus which infect the plant have ssRNA
- Viruses are facultative parasites and are inert outside their specific host cell

	(a)	(b)	(c)	(d)
(1)	T	T	T	T
(2)	T	F	F	T
(3)	F	T	T	F
(4)	F	F	T	F

- (1)
- (2)
- (3)
- (4)

161.Which of the following statements is **incorrect**?

- Water provides the medium for enzymatic activities needed for growth.
- Oxygen helps in releasing metabolic energy essential for growth activities.
- Only macronutrients are required for the synthesis of protoplasm.
- Deviation in optimum temperature range can be detrimental to the growth of plants.

162.Identify the **correct** option w.r.t. the tissue that is formed as a result of dedifferentiation.

- Cork
- Interfascicular cambium
- Secondary phloem
- Secondary cortex

163.Sporogenous tissue

- Nourishes the developing pollen grains
- Occupies the centre of each microsporangium
- Comprises of cells that undergo mitotic divisions to form microspore tetrads
- Is a vasculated sterile tissue

164.Germ pores of the pollen grain

- Are present on the inner wall of sporoderm
- Allow the pollen tube to emerge during germination
- Exhibit fascinating array of pattern and design on intine.
- Help in pollen preservation
- Are prominent apertures on the exine.

Choose the **correct** option from the following.

- (a), (b) and (d) are correct
- (b) and (e) are correct
- (c), (d) and (e) are incorrect
- (a), (b) and (c) are incorrect

165.Which of the following has become ubiquitous in occurrence and causes pollen allergy?

- Parthenium*
- Mirabilis*
- Oxalis*
- Vallisneria*

166.If starch grain size of pea is considered as the phenotype, the alleles show

- Co-dominance
- Incomplete dominance
- Complete dominance
- Pleiotropy

167.Skin colour in humans is

- An example of pleiotropy
- Seen in two distinct alternate forms
- Generally controlled by multiple genes
- The darkest when the genotype has all the recessive alleles.

The **correct** ones is/are

- (a), (b) and (d) only
- (c) only
- (b) and (c) only
- (a) and (b) only

168. Which of the following statements is **correct**?

- Initial clue about chromosomal mechanism of sex determination can be tracked back to the experiments carried out in birds.
- (1) Henking discovered X-body but could not explain the significance of X-body.
- (2) Large number of insects show XX-XY type of sex determination.
- (3) Worker honeybees are produced as a result of parthenogenesis.
- (4)

169. Read the given Assertion (A) and Reason (R) and choose the **correct** option.

Assertion (A): A recessive character that fails to get expressed in heterozygous condition may get expressed again when it becomes homozygous.

Reason (R): Segregation of characters takes place during the process of formation of zygote.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) Both (A) and (R) are false.

170. Which of the following plastids is involved in storage of proteins?

- (1) Aleuroplasts
- (2) Chloroplasts
- (3) Chromoplasts
- (4) Elaioplasts

171. Select the correct match w.r.t hormones and their functions in humans.

(1)	ADH	–	Facilitates the conversion of angiotensin I in blood to angiotensinogen
(2)	Aldosterone	–	Facilitates reabsorption of Na ⁺ and water from distal part of the renal tubule
(3)	Renin	–	Causes vasodilation
(4)	ANF	–	Increases sodium reabsorption

- (1) (1)
- (2) (2)
- (3) (3)
- (4) (4)

172. Choose the correct option w.r.t. the phases of the menstrual cycle in healthy human female in which the peak of progesterone is seen.

- (1) Menstrual phase
- (2) Ovulatory phase
- (3) Luteal phase
- (4) Proliferative phase

173. In human females, during pregnancy, increased production of certain hormones up to several folds is essential for all of the following, **except**

- (1) Supporting the foetal growth
- (2) Supporting the metabolic changes in the mother
- (3) Maintenance of regular menstrual cycle
- (4) Maintenance of pregnancy

174. Select the function of a sac-like structure in cockroach which is followed by proventriculus.

- (1) Grinding of food
- (2) Secretion of saliva
- (3) Storage of food
- (4) Storage of faeces

175. Complete the analogy by selecting the correct option w.r.t. development of foetus during pregnancy.

Formation of heart : End of 4th week of pregnancy : :
Formation of major organ systems : _____ of pregnancy

- (1) End of 6th week
- (2) End of 12th week
- (3) End of 2nd week
- (4) End of 3rd week

176. How many statement(s) given below is/are correct w.r.t humans?

- (a) There are remarkable differences between the reproductive events in the male and in the female
- (b) All reproductive events both in males and females occur only after puberty
- (c) Vas deferens leave the testis and open into epididymis
- (d) Urethra originates from penis and acts as urinogenital duct.

Choose the **correct** option.

- (1) One
- (2) Three
- (3) Two
- (4) Four

177. Select the correct route for the passage of sperms to the outside in a male frog.

- (1) Testis → Vasa efferentia → Bidder's canal → Cloaca → Urinogenital duct
- (2) Testis → Vasa efferentia → Seminal vesicle → Cloaca → Bidder's canal → Ureters
- (3) Testis → Vasa efferentia → Bidder's canal → Urinogenital duct → Cloaca
- (4) Vasa efferentia → Testis → Bidder's canal → Kidney → Ureters → Rectum

178. Which of the following is not true for condoms?

- (1) They are disposable
- (2) They give privacy to the users
- (3) They prevent users from contracting STIs
- (4) They are used as emergency contraceptives.

179. How many of the factors mentioned in the box below affect the binding of oxygen with haemoglobin?

pCO₂, H⁺ concentration, temperature, pO₂

Choose the correct option.

- (1) One
- (2) Two
- (3) Four
- (4) Three

180. Asthma is characterised by

- (1) Inflammation of bronchi and bronchioles
- (2) Damaged alveolar walls and chronic accumulation of fluid in the lungs
- (3) Proliferation of fibrous tissues in lungs
- (4) Bacterial infection resulting in inflammation in trachea



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